ASSIST09 Cheat Sheet

M6x09-II SBC — verified monitor workflow `115200,n,8,1; no handshake`

Start here Reset the board. A working monitor prints:

```
ASSIST09
>
```

Commands and numeric values are hexadecimal. Type a command in the terminal's *bottom command-entry field*, then press Enter. Direct typing is also paced, but the command-entry field is the preferred path.

Daily monitor commands

Command	Action
`R`	Display/change the saved CPU registers.
`D 1000 20`	Display `$20` bytes of memory beginning at `$1000`.
`M 1000`	Examine/change memory interactively at `$1000`.
`/`	Continue examining memory from the last memory address.
`L`	Enter S-record load mode.
`G 1000`	Start execution at `$1000`.
`C 1000`	Call the routine at `$1000` as a subroutine.
`B`	List breakpoints.
`B 1000`	Add a breakpoint at `$1000`.
`B -1000`	Remove the breakpoint at `$1000`.
`B -`	Clear all breakpoints.

Verified RAM load and run

1. At `>`, enter `L`. The prompt does *not* return while the loader is waiting.
2. In the Tcl terminal, choose **File** → **Send...** and select `src/assist-09/assist09-smoke.s19`.
3. Wait for `>` to return, then enter `G 1000`.

Expected result:

```
ASSIST09 RAM SMOKE TEST PASSED
ASSIST09
>
```

Important: use the repository Tcl terminal. It sends S-records with character pacing and terminates an ASSIST09 load using only `S9`; sending the remaining bytes of a normal `S9030000FC` record would become the next monitor input.

Other built-in commands

Cmd	Action	Cmd	Action
`P start end`	Punch memory as S-records.	`V`	Verify incoming S-records against memory.
`T count`	Trace instructions.	`.`	Trace one instruction.
`O from to`	Calculate 6809 branch offsets.	`W value`	Set the expression "window" value.
`N value`	Set output padding/nulls.	`S value`	Set stack-trace level.
`E ...`	Encode a 6809 indexed-addressing post-byte.		

Expressions and memory editing

ASSIST09 accepts hexadecimal values. In expressions, `M` means the last memory address, `P` the saved program counter, `W` the window value, and `@` an indirect value. In `M` mode, enter a hex byte to replace it; use a space for the next byte, `^` for the previous byte, and `/` to show the address again.

Board facts

Monitor ROM	`$F800`–`$FFFF`; reset vector `$F837`.
ACIA	`$BE00` Control/Status; `$BE01` Data.
Serial setup	MC6850 control `$11`: 8N1, divide-by-16, RTS low, TX interrupt disabled.
Flow control	PCB locally connects RTS to CTS. Do *not* install a CTS-to-ground jumper.

Source: `src/assist-09/assist09.asm`; operational notes: `terminal/README.md`. Advanced breakpoint, trace, punch, verify, and encode commands are present in the monitor; the load/run workflow above is the board-level accepted path.